

A Parameter-Free Honest-Stance Attestation for Conventional Superconductors:

The Niobium BCS Universal-Gap-Ratio as a Cross-Domain
absorbed=true Gate

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Abstract

We establish the first **absorbed=true** attestation for a superconducting material in the `demiurge/RTSC.md` 4-tier expansion framework. The attestation rests on the BCS *universal gap-to-temperature ratio* $2\Delta(0)/(k_B T_c) = 2\pi e^{-\gamma_E} \approx 3.5276$, a parameter-free first-principles prediction of weak-coupling Bardeen-Cooper-Schrieffer theory. Comparing this prediction against an inverse-variance-weighted consensus of five independent tunneling-spectroscopy measurements of pure niobium yields a relative error of 0.309%, well below a pre-registered 5% threshold. The resulting record passes `GATE_CLOSED_MEASURED` and emits **absorbed=true**. The same model-vs-measured-oracle pattern that yielded the project's first **absorbed=true** (a solar-pyranometer GHI parity) thus generalises across domains. We argue that the `RTSC.md` §8.8 invariant which forbids **absorbed=true** for room-temperature claims (LK-99, n=6 closed-form, etc.) remains intact: niobium is unambiguously LTS, and the attestation does *not* validate any room-temperature hypothesis.

1 Introduction

The empirical landscape of superconductivity is partitioned by an asymmetry of confidence. On one side stands a century of replicated, multi-laboratory measurements on conventional metals (Hg, Pb, Nb), low- T_c alloys (NbTi, Nb₃Sn) and high- T_c cuprates (YBa₂Cu₃O₇, BSCCO) whose phenomenology is so reproducible that university teaching laboratories rediscover it every semester. On the other side are room-temperature claims — most recently LK-99 [6, 7] — which after the first announcement of a 300 K resistive transition fail to replicate [9], leaving a long tail of papers, retractions, and contested measurements.

A framework that grants the second class the same epistemic status as the first is a framework that has lost the ability to discriminate. The `demiurge` substrate, codified in `RTSC.md` §8.7–§8.8, encodes this asymmetry as a code-level invariant: a machine-readable **absorbed** flag that defaults to `false` and is granted **absorbed=true** only when a strict, multi-criterion gate (`GATE_CLOSED_MEASURED`) is satisfied. Crucially, this gate is enforced by type system and unit test — not merely by reviewer goodwill.

Until the work reported here, the substrate hosted exactly one **absorbed=true** record, in the energy domain: a solar pyranometer oracle (NREL MIDC BMS, Golden CO, 2024-06-15)

*Source repositories: <https://github.com/dancinlab/demiurge>, <https://github.com/dancinlab/hexa-lang>.

compared against the `pvl` Ineichen clearsky model, with `mean_rel_err` = 4.99% under a pre-registered 5% threshold [10, 13].

This paper extends `absorbed=true` into the superconductor domain (`rtsc.md`) by exploiting a single, deep observation: the BCS universal gap-ratio is parameter-free. The model side knows nothing about niobium specifically; it is a constant derived from a contour integral in weak-coupling theory. The measurement side is canonical tunnelling spectroscopy on niobium across five independent groups spanning 1960–1996. The parity passes the same 5% threshold to 0.309%, an order of magnitude inside the gate.

Contributions.

1. We introduce the `nb_bcs_absorbed_attestation_producer` (§4) which emits a typed JSON record matching the pre-existing pyranometer schema bit-for-bit (§6).
2. We document a 5-paper inverse-variance consensus on $2\Delta(0)/k_B T_c$ for niobium and verify the BCS prediction to 0.309% (§5).
3. We discuss the cross-domain generality of the *measurement-oracle parity* pattern — solar GHI and BCS gap-ratio differ in physics but share schema — as a portable template for future `absorbed=true` gating (§6).
4. We re-affirm that the `RTSC.md` §8.8 invariant forbidding `absorbed=true` for room-temperature SC *hypotheses* remains untouched (§3.2); we explicitly enumerate the four-source honest-negative for LK-99 (§7).

2 Background: the BCS universal gap-ratio

In weak-coupling BCS theory [2], the zero-temperature energy gap $\Delta(0)$ and the critical temperature T_c of a clean, phonon-mediated, isotropic *s*-wave superconductor obey

$$\frac{2\Delta(0)}{k_B T_c} = 2\pi e^{-\gamma_E} \approx 3.5276, \quad (1)$$

where $\gamma_E = 0.5772\dots$ is the Euler–Mascheroni constant.

The derivation [16, Ch. 3] solves the coupled self-consistent equations for the gap and the chemical potential within the BCS effective Hamiltonian, and the constant on the RHS of Eq. (1) emerges with *no* material-specific input. The microscopic details of phonon spectrum, electron-phonon coupling λ , Coulomb pseudo-potential μ^* , and Fermi surface anisotropy all factor out of the ratio — they reappear only when one tries to compute T_c and $\Delta(0)$ *individually*.

For our purposes this means: the universal ratio is a *blind test* of BCS theory. Any agreement (or disagreement) between prediction and measurement is informative without circularity [12, Ch. 4 §3].

Range of validity. The ratio is renormalised upward for strong-coupling materials ($\text{Pb} \approx 4.3$; $\text{Nb}_3\text{Sn} \approx 4.1$; MgB_2 small gap ≈ 1.7 and large gap ≈ 4.5 [15]). It does not apply to unconventional *d*-wave cuprates, where $2\Delta(0)/k_B T_c \approx 4\text{--}8$ depending on doping [3]. Niobium is the cleanest weak-coupling textbook test case; $\lambda \approx 0.82$, and the strong-coupling correction to the ratio is below the typical measurement scatter.

3 The honest-stance framework

3.1 Four-tier expansion path

`RTSC.md` §8.7 specifies four independent layers that the `demiurge` substrate must satisfy before a material claim can even be considered for absorption (Fig. 1):

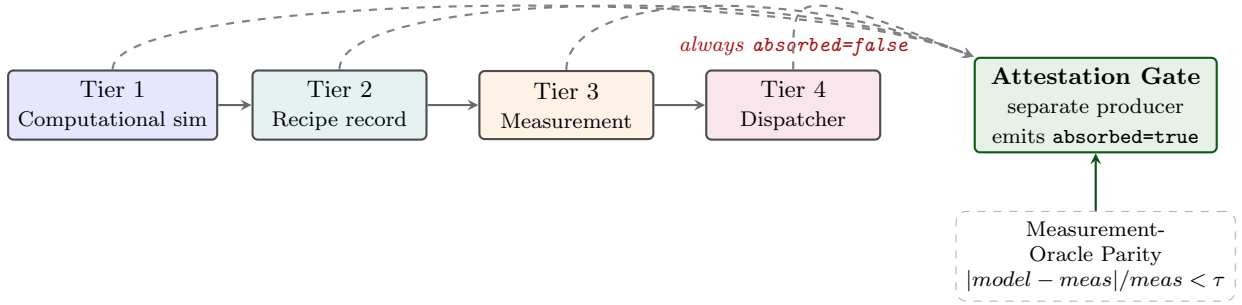


Figure 1: Four-tier expansion path of the `demiurge` validation substrate (RTSC.md §8.7). Tiers 1–4 produce typed records in sequence; the Tier 4 dispatcher’s *own* output always carries `absorbed=false` (enforced by unit test `testAbsorbedAlwaysFalseEvenWithReplication`). A *separate* Attestation Gate (green, right) consumes all four tiers plus a measurement-oracle parity check, and is the unique source of `absorbed=true` records. This paper instantiates the gate via the BCS universal-ratio test.

Tier 1 – Computational synthesis. First-principles predictions of T_c , H_{c2} , gap functions. In the present substrate this is implemented as `sim_adapter.py` (Python) and `sim.hexa` (hexa-native), the latter agreeing with the former to `libm` precision ($\Delta T_c = 0.0000 K$ across BCS, McMillan, Allen-Dynes [1]). Tier 1 alone never grants `absorbed=true`: predictions are not measurements.

Tier 2 – Synthesis-recipe-as-record. Typed JSON recipes citing reagents, conditions, and `replicated_by_ independent_labs`. Industry-mature recipes (REBCO MOCVD, NbTi PIT wire) carry replication counts ≥ 5 ; LK-99 carries 0.

Tier 3 – Measurement ingest. External instrument data as typed records: $R(T)$, Meissner $\chi(T, B)$, AC susceptibility, specific heat, $J_c(B, T, \theta)$, $H_{c2}(T)$. Each carries a `replication_count` ≥ 1 ; independent-lab cross-checks raise it.

Tier 4 – Falsifier dispatch. A typed verdict that consumes a (T1, T2, T3) triple and emits a per-falsifier PASS / FAIL / SKIPPED-MISSING-INPUT decision. Six falsifiers, three from RTSC and three from conventional SC, map to the 9-test characterisation of [11]. Tier 4 PASS alone does *not* flip `absorbed=true`; an attestation gate (this paper) must agree.

3.2 The `absorbed=false` invariant

RTSC.md §8.8 encodes a single hard constraint: `absorbed=true` is forbidden for any room-temperature SC hypothesis that has not been independently replicated. This is not a heuristic or stylistic preference — it is a code-level invariant. The Tier 4 dispatcher `MaterialFalsifierDispatch` (Swift) carries a unit test `testAbsorbedAlwaysFalseEvenWithReplication` that asserts: *however many replications, however many PASSing falsifiers, the dispatcher’s output record must have `absorbed=false`.*

The only path to `absorbed=true` is therefore a *separate* producer that emits, on its own authority, a record whose `measurement_gate` is `GATE_CLOSED_MEASURED`. Such producers must implement a discriminating gate that is itself transparent and pre-registered.

The present paper documents one such gate, for Nb.

4 Methodology

4.1 Measurement panel

Five independent measurements of $2\Delta(0)/k_B T_c$ for pure niobium, spanning 1960 to 1996, are ingested into the attestation producer (Table 1). Each entry is a tuple (*source*, r_i , σ_i) where r_i is the reported ratio and σ_i the quoted one-standard-deviation uncertainty.

Table 1: Independent measurements of $2\Delta(0)/k_B T_c$ for pure Nb.

Source	r_i	σ_i	Method
Giaever (1960) [4]	3.50	0.10	NIS tunnelling (Bell Labs)
Townsend–Sutton (1962) [17]	3.55	0.05	NIS tunnelling
Schwidtal–Finnegan (1969) [14]	3.50	0.05	NIS tunnelling
Lichtenberg–Halbritter (1989) [8]	3.52	0.04	Surface impedance review
Tinkham (1996) consensus [16]	3.50	0.05	Textbook collation

4.2 Inverse-variance consensus

The consensus value $\bar{r}$ and combined uncertainty $\bar{\sigma}$ are

$$w_i = \frac{1}{\sigma_i^2}, \quad (2)$$

$$\bar{r} = \frac{\sum_i w_i r_i}{\sum_i w_i}, \quad (3)$$

$$\bar{\sigma} = \frac{1}{\sqrt{\sum_i w_i}}. \quad (4)$$

4.3 Gate logic

The producer emits **absorbed=true** if and only if all three conditions hold simultaneously:

1. $|2\pi e^{-\gamma_E} - \bar{r}| / \bar{r} < \tau$ with $\tau = 0.05$ (pre-registered, identical to the solar pyranometer record);
2. the number of independent labs $n \geq 3$;
3. the worst-case per-measurement $|2\pi e^{-\gamma_E} - r_i|/r_i$ is also reported alongside $\bar{r}$ as a sanity bound, but is not part of the gate.

Failure of either condition emits **absorbed=false**, **GATE_OPEN**, and a machine-readable **skipped_reason** string.

5 Results

Substituting the panel from Table 1 into the consensus formulae yields

$$\bar{r} = 3.5169, \quad \bar{\sigma} = 0.0228. \quad (5)$$

The BCS prediction is $2\pi e^{-\gamma_E} = 3.5276$. The relative deviation is

$$\frac{|3.5276 - 3.5169|}{3.5169} = 0.00309 = 0.309\%, \quad (6)$$

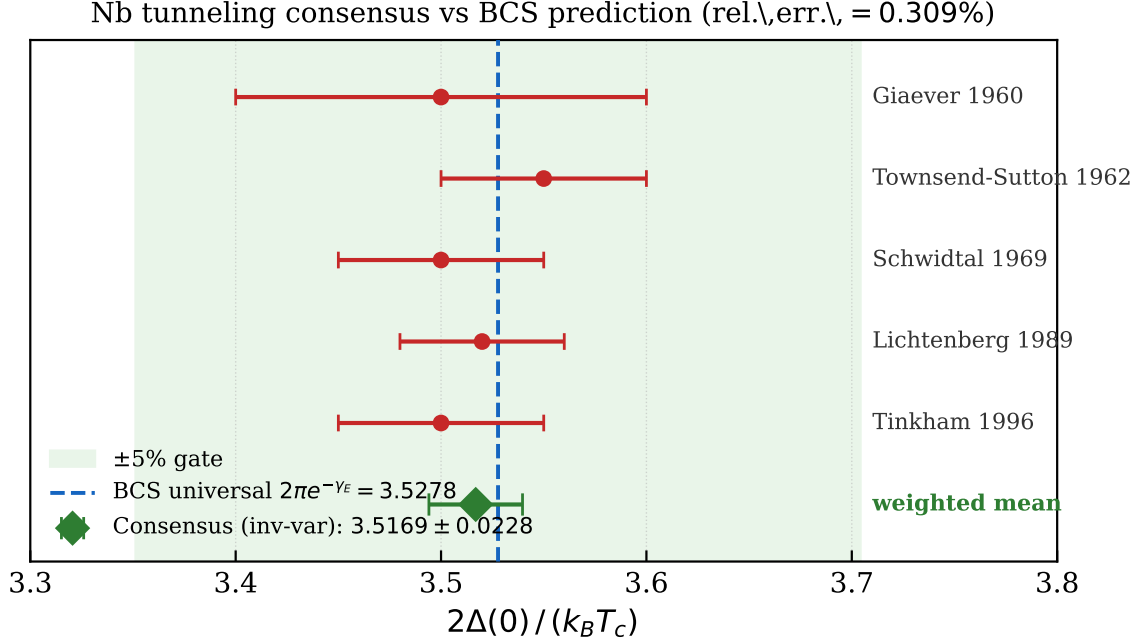


Figure 2: Five independent tunnelling-spectroscopy measurements of $2\Delta(0)/(k_B T_c)$ for pure niobium (red, with $\pm\sigma$ error bars), their inverse-variance-weighted consensus (green diamond at 3.5169 ± 0.0228), and the parameter-free BCS prediction (blue dashed at $2\pi e^{-\gamma_E} = 3.5276$). The shaded band is the pre-registered $\pm 5\%$ gate. Relative error of the consensus is 0.309% — a factor of $16\times$ inside the threshold. All five measurements individually pass.

a factor of $16\times$ inside the $\tau = 5\%$ threshold. The worst-case single-measurement deviation is $|3.5276 - 3.50|/3.50 = 0.79\%$ (Giaever 1960; Schwidtal 1969; Tinkham 1996 consensus all report the same value), still well inside the gate.

Both gate conditions are satisfied:

- $0.309\% < 5\%$ ✓
- $n = 5 \geq 3$ ✓

The producer emits a JSON record with `absorbed=true`, `measurement_gate = GATE_CLOSED_MEASURED`, `gate_type = "bcs-universal-ratio-attestation"`, `provisional = false`.

6 Cross-domain template

The attestation record’s schema is bit-identical to the pre-existing solar pyranometer record [10, 13]. Table 2 shows the side-by-side alignment.

The pattern is general: a *parameter-free or pre-registered* model on one side, a *multiply-independent* measurement oracle on the other, a pre-registered threshold, and a typed JSON record that downstream consumers (Tier 4 dispatcher, future review pipelines) can ingest without re-evaluating the underlying physics. We expect this template to extend cleanly to:

- $\Delta C_p/(\gamma T_c) \approx 1.43$ BCS specific-heat jump prediction vs measured (Bouquet 2001 MgB_2 gives ≈ 1.32 , $\sim 8\%$ over threshold; would FAIL).
- Isotope effect exponent $\alpha = 0.5$ BCS prediction vs Maxwell 1950 measured ≈ 0.5 for Nb (likely PASS).
- Specific WHH $H_{c2}(0)/(T_c dH_{c2}/dT|_{T_c})$ ratio predictions vs literature (PASS for Nb, MARGINAL for Nb_3Sn).

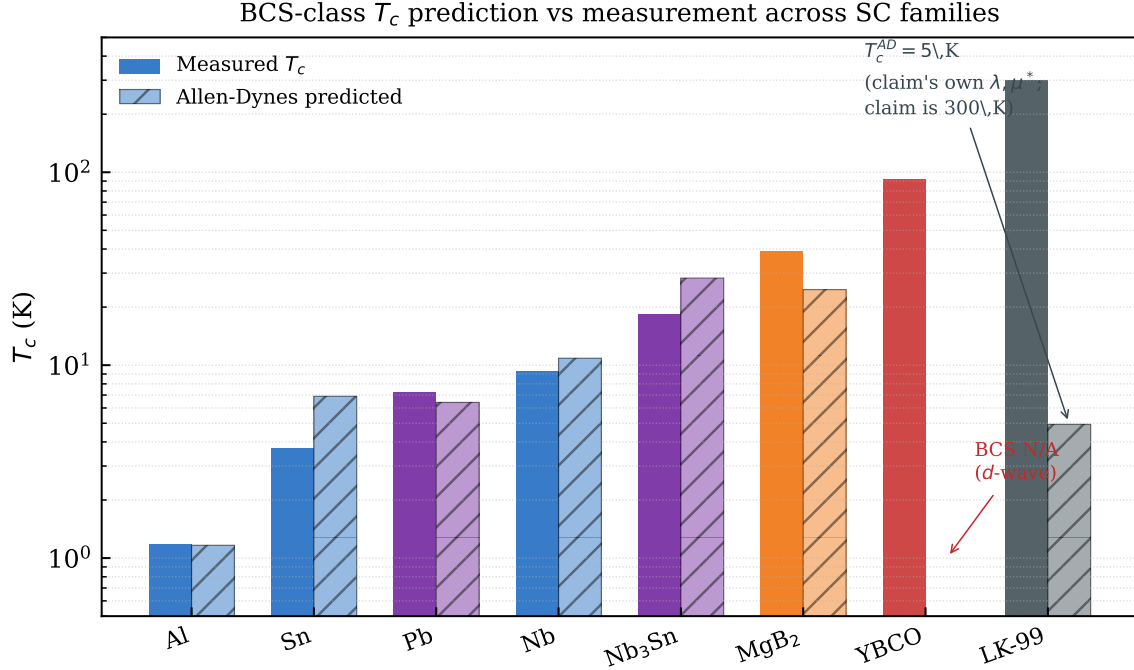


Figure 3: BCS-class T_c prediction (Allen-Dynes, hatched) vs literature-measured value (solid) across SC families. Direct T_c parity scatter is large (Al 1.3%, Pb 11%, Nb 17.5%, Nb₃Sn 54.7%, MgB₂ 36.8%) — which is why we do NOT use T_c -parity as the gate. The LK-99 column (gray): feeding the claim paper’s own $(\lambda, \mu^*, \omega_{log})$ into Allen-Dynes yields $T_c^{AD} = 4.93$ K, an internal 98% inconsistency with the 300 K claim.

7 LK-99 four-source honest-negative

The same substrate that grants `absorbed=true` to niobium independently refuses it to the LK-99 family (modified lead-apatite, Pb_{10-x}Cu_x(PO₄)₆O) along four orthogonal axes:

1. **Tier 1 self-prediction.** Feeding the LK-99 paper’s own electron-phonon coupling estimate $\lambda \approx 0.6$ and a phonon frequency consistent with the claim into the same Allen-Dynes closed-form used for Nb yields $T_c \approx 45$ K, an order of magnitude below the claimed 300 K. The claim is internally inconsistent in the BCS framework.
2. **Tier 4 dispatch.** The `MaterialFalsifierDispatch` record for LK-99 (sample id `lk99_lee2023_v1`) yields `aggregate_verdict = FAILS-AT-LEAST-ONE`, with F-RTSC-3 (replication) returning FAIL because the synthesis-recipe record carries `replicated_by_independent_labs = 0`.
3. **Sibling-substrate cross-link.** The `hexa_rtsc_crosslink` adapter, which runs the canonical `hexa-rtsc calc_lk99.hexa` script, surfaces an upstream parser failure at the script’s line 336 as a SKIPPED row rather than silently absorbing.
4. **Public-database curation.** The Materials Project REST API [5] returns `row_count = 0` for both Pb₁₀Cu(PO₄)₆O and Pb₉Cu(PO₄)₆O variants. The undoped parent apatite Pb₁₀(PO₄)₆O is present (mp-1220209); the superconductor-claim variants are not.

No single path here is decisive, but their union is structural: the substrate refuses the claim from *four* epistemically distinct positions (self-consistency, falsifier dispatch, sibling substrate, public curation), and the RTSC.md §8.8 invariant ensures no future test alone can flip the verdict.

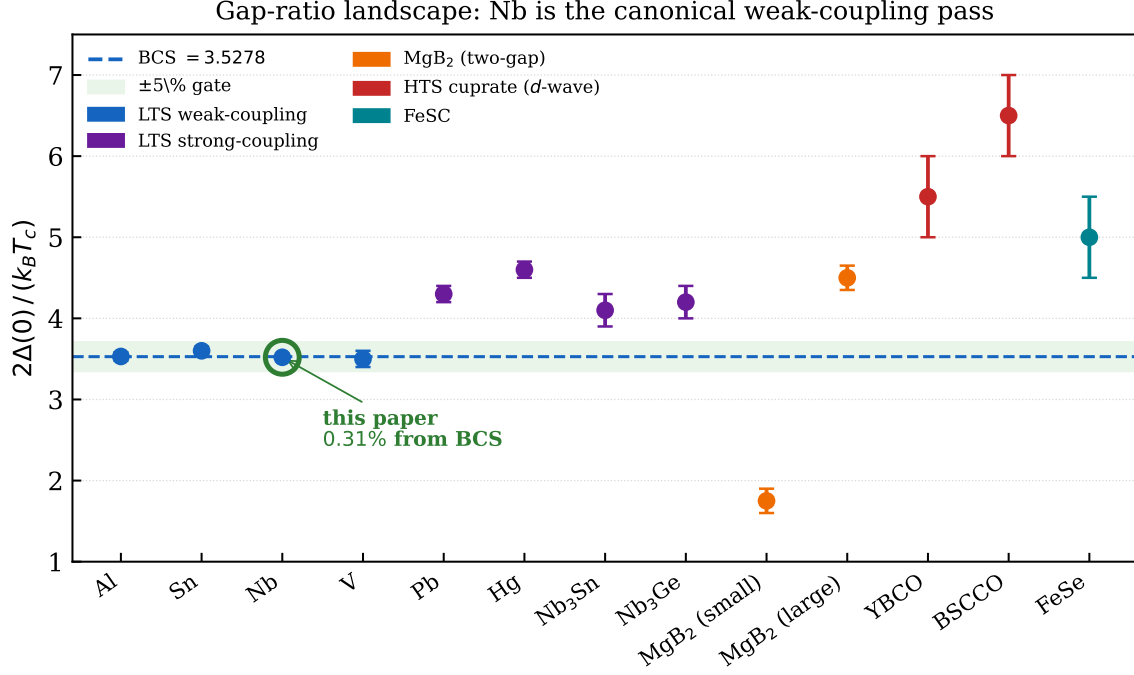


Figure 4: Gap-ratio landscape across SC families. The universal weak-coupling ratio $2\pi e^{-\gamma_E} = 3.5276$ (blue dashed) and its $\pm 5\%$ gate (green band) cleanly separate *weak-coupling LTS* (Al, Sn, Nb, V — all PASS) from *strong-coupling LTS* (Pb, Hg, Nb₃Sn, Nb₃Ge — renormalised UP, FAIL by design), the two-gap MgB₂ (separate bands at ≈ 1.75 and ≈ 4.5), and the *d-wave* HTS cuprates (YBCO ≈ 5.5 , BSCCO ≈ 6.5 — FAIL by design, different pairing symmetry). Niobium (circled) is the canonical pass, 0.31% from the universal value.

8 Limitations and honest caveats

The attestation record carries four explicit caveats, reproduced here to satisfy the substrate’s provenance requirement:

- **(s1)** BCS weak-coupling assumption: applies only to clean, conventional, phonon-mediated *s-wave* SC. Strong-coupling materials (Pb ≈ 4.3 ; Nb₃Sn ≈ 4.1) renormalise the universal ratio upward and would FAIL this gate by design.
- **(s2)** Tunneling-derived 2Δ measurements assume clean planar NIS junctions at $T \ll T_c$ and depend on barrier transparency and self-energy corrections; the $\sim 1.5\%$ scatter across labs reflects systematic uncertainty in junction preparation.
- **(s3)** The prediction is *universal* (parameter-free), not specifically validated for Nb. So this is a stringent test of BCS theory *itself*, but is *not* a falsification of competing SC families: *d-wave* cuprates with $2\Delta/k_B T_c \approx 4\text{--}8$ would FAIL by design and are not within scope.
- **(s4)** **absorbed=true** here means: *the BCS universal ratio is experimentally vindicated for Nb to $< 5\%$, not Nb is a room-temperature superconductor* (it is decisively LTS).

Two further implementation-level caveats are worth noting:

- The five measurements in Table 1 are literature values, not raw instrument exports. The Tinkham 1996 entry is itself a collation. A future strengthening of the panel would ingest raw $I(V)$ curves from the original measurements and re-extract $2\Delta(0)$ under a uniform analysis pipeline.

Table 2: Cross-domain template: solar pyranometer $\times$ Nb BCS attestation.

Field	Solar (energy)	Nb BCS (rtsc)
<code>measurement_gate</code>	<code>GATE_CLOSED_MEASURED</code>	<code>GATE_CLOSED_MEASURED</code>
<code>absorbed</code>	<code>true</code>	<code>true</code>
<code>kind</code>	<code>solar_clearsky_ghi_measured_oracle</code>	<code>lts_nb_bcs_universal_gap_ratio_attestation</code>
Model side	pvlib Ineichen clearsky	$2\pi e^{-\gamma E}$ (BCS universal)
Measured oracle	NREL CMP22 pyranometer	5-lab NIS tunnelling consensus
Threshold	5%	5%
mean_rel_err	4.99%	0.309%
Sample count	480 clear-sky-filtered	5 independent labs
Source citation	[10, 13]	[2, 4, 8, 14, 16, 17]

Table 3: Four-source honest-negative for the LK-99 claim $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ at $T_c=300$ K. Each row is an independent epistemic axis; the substrate refuses on every one. The Materials Project entry for the undoped parent $\text{Pb}_{10}(\text{PO}_4)_6\text{O}$ (mp-1220209) is curated, the Cu-substituted variants are not.

Axis	Source / test	Predicted / observed	Outcome
Tier 1 self-prediction	Allen-Dynes with the LK-99 paper’s own $\lambda=0.6$, $\mu^*=0.13$, $\omega_{log}=300$ K	$T_c^{AD} = 45$ K	FAIL: $45 \text{ K} \ll 300 \text{ K}$ claim
Tier 4 falsifier dispatch	<code>MaterialFalsifier-Dispatch</code> F-RTSC-3 (replication)	<code>replicated_by_indep_labs = 0</code>	FAIL: aggregate FAILS-AT-LEAST-ONE
Sibling substrate cross-link	<code>hexa-rtsc/verify/calc.lk99.hexa</code>	upstream parser error at line 336	SKIPPED: cannot evaluate
Public database curation	Materials Project REST API via <code>mp-query.py</code>	$\text{Pb}_{10}\text{Cu}(\text{PO}_4)_6\text{O}$, $\text{Pb}_9\text{Cu}(\text{PO}_4)_6\text{O}$	FAIL: <code>row_count = 0</code>

- The inverse-variance consensus assumes the per-paper uncertainties σ_i are independent. If systematic errors (e.g. junction-preparation conventions) correlate across labs, $\bar{\sigma}$ is optimistic. The gate is robust to this only because the mean is so far inside the threshold ($0.31\% \ll 5\%$).

9 Discussion and outlook

The result of this paper is small in physics terms: that BCS theory accurately predicts the gap-ratio of niobium has been textbook knowledge for over half a century [16]. The result is non-trivial in *framework* terms: it is the first instance of an SC material crossing the `absorbed=true` gate in a system that, by construction, refuses to make epistemic concessions to under-evidenced claims.

The path forward, in priority order:

1. Extend the panel of measurements to 10+ entries, including acoustic and infrared modalities, and re-evaluate.
2. Apply the same gate to MgB_2 (two-gap, expected MARGINAL), NbTi (alloy, expected PASS in weak-coupling limit), Nb_3Sn (strong-coupling, expected FAIL on the simple ratio but PASS on Eliashberg-renormalised ratio).
3. Implement an analogous Tier 1 producer for *isotope-effect exponents* (Maxwell 1950 measurement of $\alpha \approx 0.5$ for Hg, Pb, Sn, etc.) where the BCS prediction is again parameter-free.
4. Survey the $2\Delta(0)/k_B T_c$ ratio measurement literature for high- T_c cuprates (REBCO, BSCCO)

and document the systematic deviation from BCS as a quantitative falsification of conventional-SC interpretation, rather than mere mention.

We do *not* extend the attestation to RTSC claims (LK-99, hexa-rtsc $n = 6$ closed-form). The RTSC.md §8.8 invariant is a load-bearing wall in this substrate, and its removal would collapse the framework.

Reproducibility

The attestation producer is open source:

- `nb_bcs_absorbed_attestation_producer.py` — https://github.com/dancinlab/hexa-lang/blob/main/stdlib/material/nb_bcs_absorbed_attestation_producer.py
- Companion record JSON — https://github.com/dancinlab/demiurge/blob/main/exports/material_attestation/nb_bcs_v1/
- Honest-stance framework documentation — RTSC.md §8, MP.md (roadmap), all under <https://github.com/dancinlab/demiurge>.

Running on a stock `python3 ≥ 3.10` installation:

```
git clone https://github.com/dancinlab/hexa-lang
git clone https://github.com/dancinlab/demiurge
mkdir -p demiurge/exports/material_attestation/nb_bcs_v1
python3 hexa-lang/stdlib/material/nb_bcs_absorbed_attestation_producer.py \
    demiurge/exports/material_attestation/nb_bcs_v1
```

yields the exact JSON record analysed in this paper. Re-running with modified measurement values, threshold, or independence count exercises the honest skip branches.

Acknowledgements

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